

Name:

Maths Class:

SYDNEY TECHNICAL HIGH SCHOOL



Year 12 Mathematics

HSC Course

Assessment 3

June, 2015

Time allowed: 90 minutes

General Instructions:

- Marks for each question are indicated on the question.
- Approved calculators may be used
- All necessary working should be shown
- Full marks may not be awarded for careless work or illegible writing
- **Begin each question on a new page**
- Write using black or blue pen
- All answers are to be in the writing booklet provided
- A set of Standard Integrals is provided at the rear of this Question Booklet, and may be removed at any time.

Section 1 Multiple Choice
Questions 1-10
10 Marks

Section II Questions 11-15
48 Marks

Section 1
Total Marks – 5
Attempt 1-5

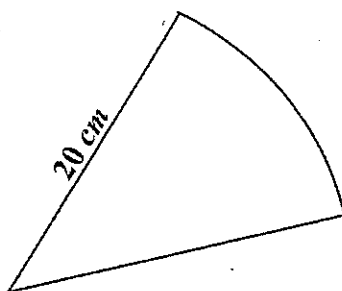
Objective response Questions

Answer each question on the multiple choice sheet provided

1. Which term represents the distance that $y = a \sin(bx)$ extends out from the centre of its graph on the y-axis?
- (A) Amplitude
(B) Domain
(C) Period
(D) Range
2. What are the solutions of $2 \cos x = -\sqrt{3}$ for $0 \leq x \leq 2\pi$?
- (A) $\frac{\pi}{6}$ and $\frac{\pi}{6}$
(B) $\frac{5\pi}{6}$ and $\frac{7\pi}{6}$
(C) $\frac{\pi}{3}$ and $\frac{2\pi}{3}$
(D) $\frac{\pi}{6}$ and $\frac{7\pi}{6}$
3. What is the derivative of $e^x(x^2 + 2x)$?
- (A) $(2x + 2)$
(B) $e^x(2x + 2)$
(C) $e^x(x^2 - 2)$
(D) $e^x(x^2 + 4x + 2)$

4. A chord of length 5 cm is drawn in a circle of radius 6 cm. The area of the smaller region inside the circle cut off by the chord, correct to one decimal place, is:
- (A) 1.8 cm^2
 (B) 2.3 cm^2
 (C) 13.6 cm^2
 (D) 15.5 cm^2

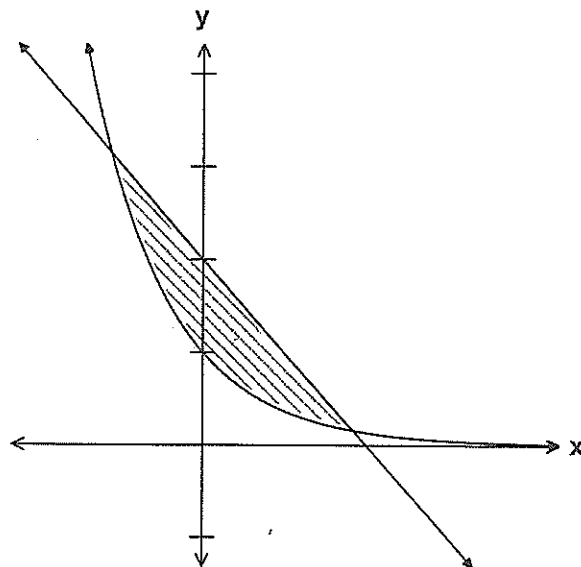
5. What is the perimeter, P , of the sector below with an angle 36° and radius 20cm?



- (A) $P = 0.5 \times 400 \times \left(\frac{\pi}{5} - \sin \frac{\pi}{5} \right) \text{ cm}$
 (B) $P = \left(0.5 \times 400 \times \frac{\pi}{5} \right) \text{ cm}$
 (C) $P = (40 + 36^\circ) \text{ cm}$
 (D) $P = (40 + 4\pi) \text{ cm}$
6. What is the value of $\int_0^1 (e^{3x} - 1) dx$?

- (A) $\frac{e^3}{3}$
 (B) $\frac{e^3}{3} - 1$
 (C) $e^3 - 1$
 (D) $\frac{1}{3} (e^3 - 4)$

7. The diagram shows the region enclosed by $x + y = 2$ and $y = e^{-x}$



Which of the following pair of inequalities describes the shaded region in the diagram?

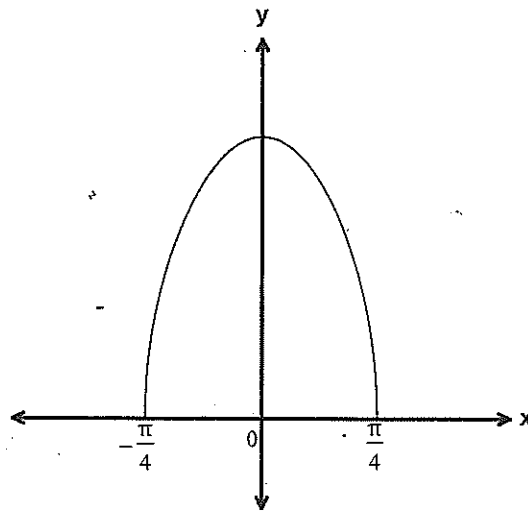
- (A) $x + y \leq 2$ and $y \leq e^{-x}$
- (B) $x + y \leq 2$ and $y \geq e^{-x}$
- (C) $x + y \geq 2$ and $y \leq e^{-x}$
- (D) $x + y \geq 2$ and $y \geq e^{-x}$
8. What is the greatest value taken by the function $f(x) = 4 - 2\cos x$ for $x \geq 0$?
- (A) 2
- (B) 4
- (C) 6
- (D) 8

9. The values for a continuous function are given in the table below.

x	0	1	2	3	4	5	6	7	8
$f(x)$	15	12.5	6	-3	-5	2	3.5	7.5	10

The trapezoidal rule approximation for $\int_0^8 f(x) dx$ is:

- (A) 36
 (B) 35.5
 (C) 48.5
 (D) 49
10. The diagram below shows the region bounded by the curve $y = \sqrt{5\cos^2 x}$ and the x-axis for $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$. The region is rotated about the x-axis to form a solid. Which of the following gives the volume of the solid?



- (A) $V = 5\pi \int_0^{\pi/4} \cos^2 x dx$
 (B) $V = 10\pi \int_0^{\pi/4} \cos^2 x dx$
 (C) $V = 10\pi \int_0^{\pi/4} \cos^4 x dx$
 (D) $V = 25\pi \int_0^{\pi/4} \cos^2 x dx$

Section II**Total Marks 50****Attempt Questions 11-15**

Answer the questions in the booklet provide. Start each question on a NEW sheet of paper.

Question 11**10 marks**

- a) Find the exact value of $\tan \frac{2\pi}{3}$ 2
- b) (i) Find the derivative of $y = \sin^2 x$ 2
- (ii) Find the equation of the tangent to $y = \sin^2 x$ at $x = \frac{\pi}{4}$ 2
- (iii) Find the equation of the normal to $y = \sin^2 x$ at $x = \frac{\pi}{4}$ 2
- (iv) If the tangent meets the x-axis at P and the normal meets the y-axis at Q, find the area of $\triangle OPQ$ where O is the origin in exact form. 2

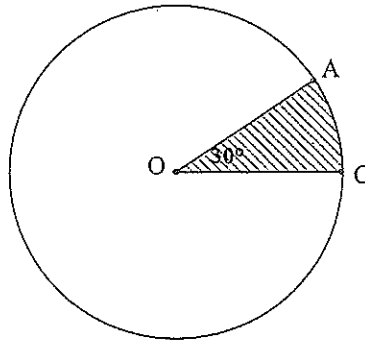
Question 12

(Start a New Page)

10 marks

- a) Differentiate $4 \cos(5x - 3)$ with respect to x :

2



- b) (i) Find the radius of the circle if the area of the shaded sector is $12\pi \text{ cm}^2$

3

- (ii) Hence find the exact length of the major arc AC

2

- c) Copy the table of values into your writing booklet and supply the missing numbers, for $f(x) = x \sin x$, writing each correct to 3 decimal places.

x	1	1.5	2	2.5	3
$f(x) = x \sin x$	0.841				

Use Simpson's Rule with 5 function values to find an approximation for $\int_1^3 x \sin x \, dx$

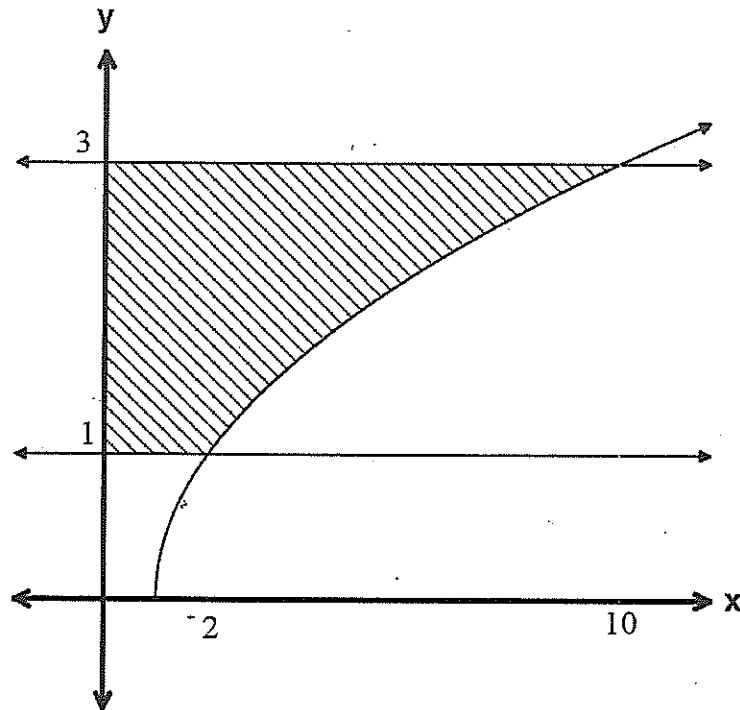
3

Question 13

(Start a New Page)

8 marks

- a) Differentiate $\frac{x}{\cos x}$ 2
- b) Find the equation of the tangent to the curve $y = 3e^x - 1$ at the point where $x = 1$ 3
- c) The diagram shows the shaded region enclosed by the curve $y = \sqrt{x-1}$, the y-axis and the lines $y=1$ and $y=3$



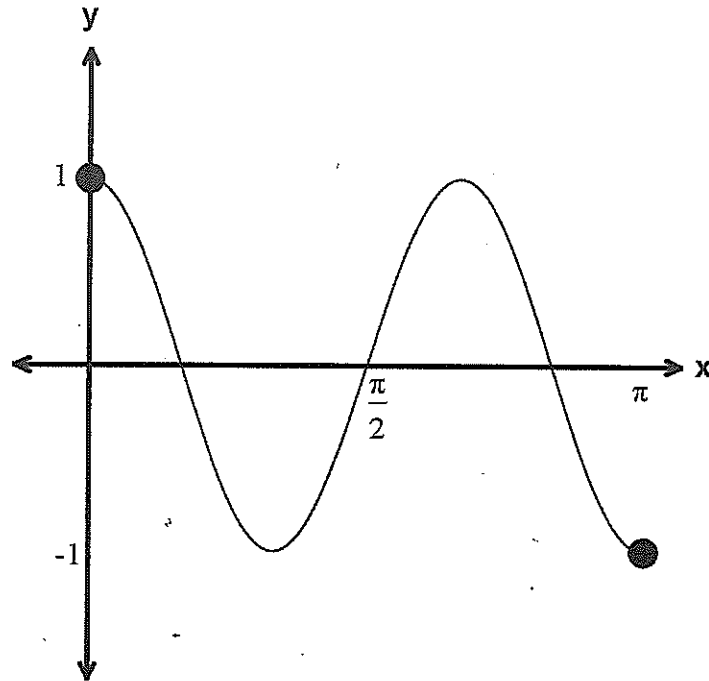
Find the volume of the solid of revolution when the shaded region is rotated about the y-axis.

3

a) Find $\int_0^{\frac{\pi}{12}} \sec^2 3x \, dx$

2

b) The graph of $y = \cos 3x$ is shown below



- (i) Solve $\cos 3x = 0$ for $0 \leq x \leq \pi$ 2
- (ii) State the amplitude and the period of $y = \cos 3x$ 2
- (iii) Copy this diagram into your booklet showing the x-intercepts
Hence sketch the graph of $y = \sec 3x$ in the domain $0 \leq x \leq \pi$
showing any asymptotes. 2
(Hint: The diagram should be one third of your page, use a ruler)
- (iv) Using (iii), find the number of solutions to $\sec 3x = x$ in the domain $0 \leq x \leq \pi$ 2

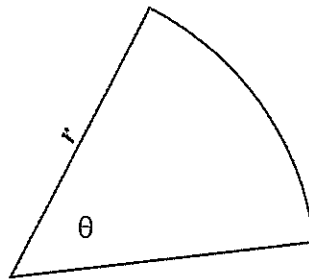
Question 15

(Start a New Page)

10 marks

- a) Consider the function $f(x) = \cos^2 x - \sin x$ in the domain $\pi \leq x \leq \frac{3\pi}{2}$
- (i) Find $f'(x)$. 1
- (ii) Find the x-coordinates of the stationary points of $y = f(x)$ and determine their nature 3

- b) The diagram shows a sector of a circle with radius r cm. The angle at the centre is θ radians and the area is 18 cm^2



- i) Find an expression for r in terms of θ . 1
- ii) Show that P , the perimeter of the sector in cm, is given by 2

$$P = \frac{6(2 + \theta)}{\sqrt{\theta}}$$

- iii) Find the minimum perimeter and the value of θ for which this occurs. 3

End of Exam



Multiple Choice

- | | |
|-----|------|
| 1 A | 6 D |
| 2 B | 7 B |
| 3 D | 8 C |
| 4 A | 9 A |
| 5 D | 10 B |

Question 11

a) $\tan \frac{2\pi}{3} = -\tan \frac{\pi}{3}$
 $= -\sqrt{3}$

b) $y = \sin^2 x$
 Let $u = \sin x$
 $du = \cos x$
 $y = u^2$
 $\frac{dy}{dx} = 2u$
 $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$
 $= 2u \times \cos x$
 $= 2 \sin x \cos x$

c) When $x = \frac{\pi}{4}$ $y = \frac{1}{2}$
 Equation of the tangent
 $\frac{dy}{dx} = 2 \times \sin \frac{\pi}{4} \times \cos \frac{\pi}{4}$
 $= 2 \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}$
 $= 1$

$y - y_1 = m(x - x_1)$
 $y - \frac{1}{2} = 1(x - \frac{\pi}{4})$
 $y = x + \frac{1}{2} - \frac{\pi}{4}$

d) Equation of the normal
 $m_2 = -1$

$y - y_1 = m(x - x_1)$
 $y - \frac{1}{2} = -1(x - \frac{\pi}{4})$
 $y = -x + \frac{\pi}{4} + \frac{1}{2}$

e) $Q(0, \frac{\pi}{4} + \frac{1}{2})$
 $P(\frac{1}{2} - \frac{\pi}{4}, 0)$

Area of $\Delta OPQ =$
 $= \frac{1}{2}(\frac{\pi}{4} - \frac{1}{2})(\frac{\pi}{4} + \frac{1}{2})$
 $= \frac{1}{2}(\frac{\pi^2}{16} - \frac{1}{4})$
 $= (\frac{\pi^2}{32} - \frac{1}{8}) \text{ units}^2$

Question 12

a) $y = 4 \cos(5x - 3)$
 $\frac{dy}{dx} = -20 \sin(5x - 3)$

b) i) Area of sector $= \frac{1}{2} r^2 \theta$
 $12\pi = \frac{1}{2} \times r^2 \times \frac{\pi}{6}$
 $144\pi = r^2 \pi$
 $r = 12$

ii) Major arc $= l = r\theta$
 $\theta = \frac{11\pi}{6}$ $L = 12 \times \frac{11\pi}{6}$
 $l = 22\pi \text{ units}$

x	1	1.5	2	2.5	3
f(x)	0.841	1.496	1.819	1.496	0.423

c) $\int_{0.5}^{3} x \sin x dx =$
 $= \frac{1}{3} [0.841 + 0.423 + 4(1.496 + 1.496) + 2(1.819)]$
 $= 2.812 \text{ (3 d p)}$

Question 13

a) $u = x$ $v = \cos x$
 $du = 1$ $dv = -\sin x$
 $\frac{dy}{dx} = \frac{\cos x \times 1 + x \sin x}{\cos^2 x}$

b) $y = 3e^x - 1$
 $\frac{dy}{dx} = 3e^x$
 When $x = 1$ $y = 3e - 1$
 $m = 3e$
 Equation of tangent
 $y - (3e - 1) = 3e(x - 1)$
 $y - 3e + 1 = 3ex - 3e$
 $y + 1 = 3ex$
 $y = 3xe - 1$

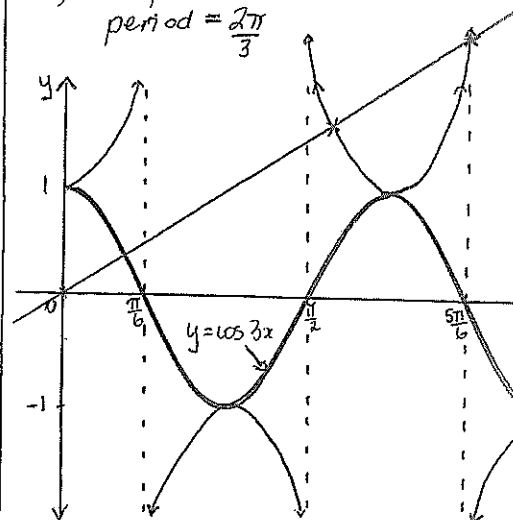
c) $y = \sqrt{x-1}$
 $y^2 = x-1$
 $x = y^2 + 1$
 $V = \pi \int_1^3 (y^2 + 1)^2 dy$
 $= \pi \int_1^3 y^4 + 2y^2 + 1 dy$
 $= \pi \left[\frac{y^5}{5} + \frac{2y^3}{3} + y \right]_1^3$
 $= \pi \left[\frac{243}{5} + \frac{54}{3} + 3 - \left(\frac{1}{5} + \frac{2}{3} + 1 \right) \right]$
 $= \pi \left[\frac{348}{5} - \frac{28}{15} \right]$
 $= \frac{1016\pi}{15} \text{ unit}^3$

Question 14

a) $\int_0^{\pi/12} \sec^2 3x dx = \left[\frac{\tan 3x}{3} \right]_0^{\pi/12}$
 $= \left[\frac{1}{3} - 0 \right]$
 $= \frac{1}{3}$

b) i) $\cos 3x = 0$
 $3x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}$
 $x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}$

ii) amp = 1
 period = $\frac{2\pi}{3}$



iv) $\sec 3x = x$
 2 solutions from the graph

Question 15

$$f(x) = \cos^2 x - \sin x$$

$$f'(x) = 2 \cos x (-\sin x) - \cos x$$

$$= -2 \sin x \cos x - \cos x$$

$$=$$

i) Stationary points occur when

$$f'(x) = 0$$

$$-2 \sin x \cos x - \cos x = 0$$

$$-\cos x (2 \sin x + 1) = 0$$

$$-\cos x = 0 \quad 2 \sin x + 1 = 0$$

$$x = \frac{\pi}{2}, \frac{3\pi}{2} \quad \sin x = -\frac{1}{2}$$

$$x = \frac{7\pi}{6}$$

within the domain

3	$\frac{7\pi}{6}$	4	$\frac{3\pi}{2}$	5
1.26	0	-0.34	0	0.26
/	-	\	-	/

$f(x)$ has a maximum at $x = \frac{7\pi}{6}$

$f(x)$ has a minimum at $x = \frac{3\pi}{2}$

b) i) $A = \frac{1}{2} r^2 \theta$

$$18 = \frac{1}{2} \theta r^2$$

$$36 = r^2 \theta$$

$$r = \sqrt{\frac{36}{\theta}}$$

$$r = \frac{6}{\sqrt{\theta}}$$

ii) $P = 2r + r \theta$

$$= 2 \times \frac{6}{\sqrt{\theta}} + \frac{6}{\sqrt{\theta}} \times \theta$$

$$= \frac{6(2 + \theta)}{\sqrt{\theta}}$$

iii) $P = 12\theta^{-\frac{1}{2}} + 6\theta^{\frac{1}{2}}$

$$P' = -6\theta^{-\frac{3}{2}} + 3\theta^{-\frac{1}{2}}$$

Stationary points occur when $P' = 0$

$$0 = -6\theta^{-\frac{3}{2}} + 3\theta^{-\frac{1}{2}}$$

$$6\theta^{-\frac{3}{2}} = 3\theta^{-\frac{1}{2}}$$

$$\frac{6}{\theta^{\frac{3}{2}}} = \frac{3}{\theta^{\frac{1}{2}}}$$

$$6\theta^{\frac{1}{2}} = 3\theta^{\frac{3}{2}}$$

$$0 = \theta^{\frac{1}{2}}(\theta - 2) \quad \theta \neq 0$$

$$\theta = 2$$

$$P'' = 9\theta^{-\frac{5}{2}} - \frac{3}{2}\theta^{-\frac{3}{2}}$$

At $\theta = 2$

$$P'' = 9(2)^{-\frac{5}{2}} - \frac{3}{2}(2)^{-\frac{3}{2}}$$

$$= 1.06 > 0 \quad \text{concave up}$$

$\therefore$ minimum at $\theta = 2$

$$P = \frac{6(4)}{\sqrt{2}}$$

$$P = 12\sqrt{2} \text{ cm}$$